

STAT 251 — Homework 4

James Bruce

```
# Global setup
set.seed(123) # for reproducibility
digits <- 3   # inline values round to 3 digits
```

Question 1

The built-in faithful dataset records 272 eruptions of the Old Faithful geyser: the duration of each eruption and the waiting time until the next one (both in minutes).

(a)

Report the mean and standard deviation of the waiting times.

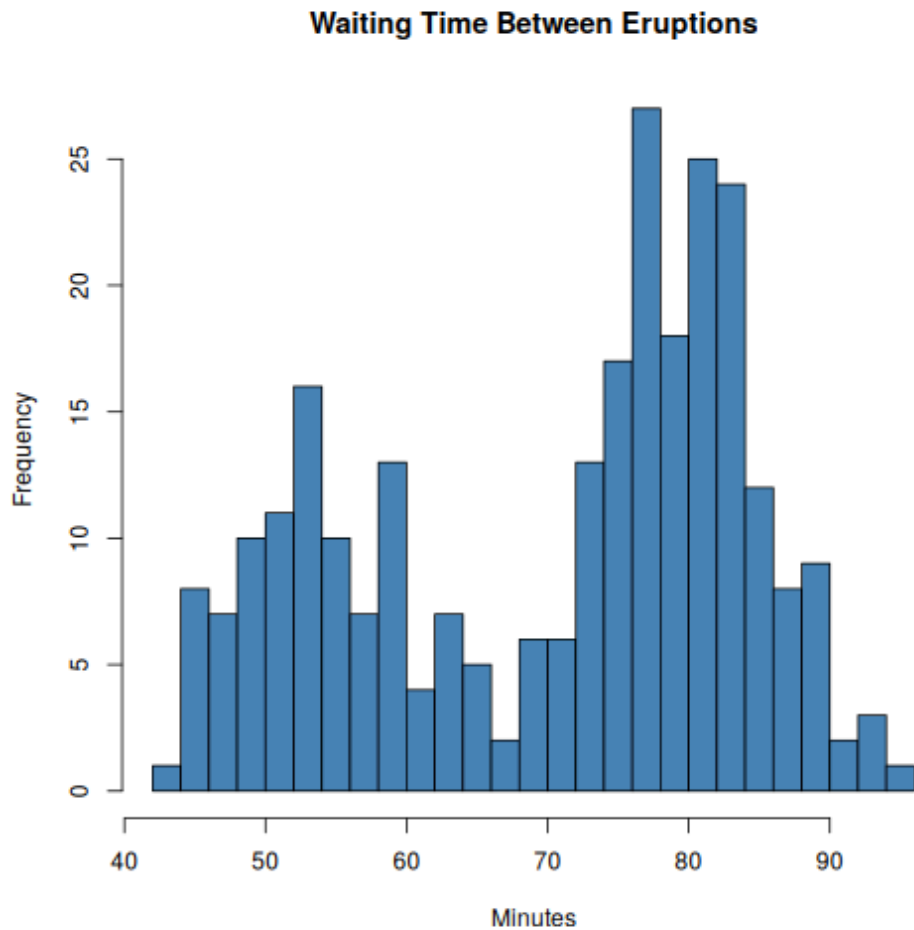
```
wait <- faithful$waiting
n <- length(wait)
```

Across the 272 recorded eruptions, the mean waiting time is 70.897 minutes with a standard deviation of 13.595 minutes.

(b)

Plot a histogram of the waiting times. What do you notice?

```
hist(wait,
      breaks = 20,
      main = "Waiting Time Between Eruptions",
      xlab = "Minutes",
      col = "steelblue")
```



The distribution is clearly *bimodal*: eruptions cluster around short (about 55 min) and long (about 80 min) waits, so the mean alone is a poor summary of a “typical” wait.

Question 2

Suppose 65% of eruptions are “long” (waiting time above 68 minutes). If a visitor watches 8 independent eruptions, what is the probability that exactly 6 of them are long?

Let X be the number of long eruptions. Then $X \sim \text{Binomial}(8, 0.65)$, so

$$P(X = 6) = \binom{8}{6} (0.65)^6 (0.35)^2 \quad (1)$$

```
p6 <- dbinom(6, size = 8, prob = 0.65)
```

Evaluating Equation 1 gives $P(X = 6) = 0.259$.

Question 3

Construct a 95% confidence interval for the true mean waiting time.

With $n = 272$ observations, the interval is

$$\bar{x} \pm t_{n-1}^* \frac{s}{\sqrt{n}} \quad (2)$$

```
ci <- t.test(wait)$conf.int # results='hide': compute now, report inline
```

which evaluates to (69.274, 72.52) minutes. We are 95% confident the true mean waiting time lies in this interval — though given the bimodality from Question 1b, the *mean* wait is not a wait any visitor is likely to experience.

Question 4

Fit the simple linear regression of waiting time on eruption duration. State the model, report the fitted slope, and assess its significance.

The model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \text{Normal}(0, \sigma^2) \quad (3)$$

where Y_i is the waiting time and x_i the duration of eruption i .

```
fit <- lm(waiting ~ eruptions, data = faithful)
slope <- coef(fit)[2]
pval <- summary(fit)$coefficients[2, 4]

plot(faithful$eruptions, faithful$waiting,
     main = "Waiting Time vs. Eruption Duration",
     xlab = "Eruption duration (min)",
     ylab = "Waiting time (min)",
     pch = 19, col = "steelblue")
abline(fit, lwd = 2)
```



Each additional minute of eruption predicts a 10.73-minute longer wait. The slope is overwhelmingly significant ($p = 8.13 \times 10^{-100}$ – tiny inline values are rendered automatically in scientific notation), which matches the strong linear trend visible in the plot.

Appendix: tweave shorthand cheat sheet

All of these come from the `tweave` Typst package and work in any math block:

Write	Get	Meaning
<code>X iid Normal(mu, sigma^2)</code>	$X \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$	i.i.d. statement; Normal, Poisson, Binomial, ... render upright
<code>H0: mu = 0</code>	$H_0 : \mu = 0$	hypotheses: H0 and HA (any casing)
<code>bar(x)</code>	$\bar{x}$	sample mean (overline, not x)
<code>choose(n, k)</code>	$\binom{n}{k}$	binomial coefficient (alias of binom)

Write	Get	Meaning
<code>integral x f(x) dx</code>	$\int x f(x) dx$	dx, dy, dz, du, dv, dw, dtheta render as differentials

For question numbering, prompt boxes, and other homework structure, tweave composes with anything: a homework template from Typst Universe, or a few #let definitions of your own (like prompt at the top of this file).